

General Characteristics

121. On the basis of position in the electrochemical series, the metal which does not displace hydrogen from water and acid is
 (a) *Hg* (b) *Al*
 (c) *Pb* (d) *Ba*
122. The tendency of $3d$ -metal ions to form stable complexes is due to their
 (a) Variable oxidation state
 (b) Strong electronegative nature
 (c) High charge/size ratio and vacant d -orbitals
 (d) Very low ionization energies
123. The $3d$ -metal ions are paramagnetic in nature because
 (a) They are reducing agents
 (b) They form coloured salts
 (c) They have one or more paired s -electrons
 (d) They have one or more unpaired d -electrons
124. Lanthanide contraction occurs because
 (a) f -orbital electrons are easily lost
 (b) f -orbital an incompletely filled
 (c) f -orbital electron an poor shielders of nuclear charge
 (d) f -orbital do not come out on the surface of atom and are buried inside
125. Which is most reactive metal
 (a) *Fe* (b) *Pt*
 (c) *Ni* (d) *Co*
126. Least reactive metal is
 (a) *Fe* (b) *Os*
 (c) *Ni* (d) *Pt*
127. Which occludes hydrogen
 (a) *Os* (b) *Pt*
 (c) *Ni* (d) All of these
128. Which has the maximum ferromagnetic character
 (a) *Fe* (b) *Co*
 (c) *Ni* (d) *Pt*
129. Which forms interstitial compounds
 (a) *Fe* (b) *Co*
 (c) *Ni* (d) All of these
130. Which occurs in nature in free state
 (a) *Fe* (b) *Co*
 (c) *Ni* (d) *Pt*
131. $3d^{10}4s^0$ electronic configuration exhibits
 (a) Zn^{++} (b) Cu^{++}
 (c) Cd^{++} (d) Hg^{++}
132. $3d^04s^0$ electronic configuration exhibits
 (a) Pd^{+2} (b) Sc^{+2}
 (c) Ti^{+4} (d) Zn^{+2}
133. Rare-earth elements are exhibited by
 (a) At. No. 58 to 71
 (b) At. No. 21 to 30
 (c) At. No. 39 to 71
 (d) At. No. 81 to 91



134. All those elements belong to *f* - block whose atomic numbers are
 (a) 58 to 71 (b) 90 to 103
 (c) Both (a) and (b) (d) None
135. The correct order of density is
 (a) $Cu > Ni > Zn > Sc$
 (b) $Ni > Cu > Zn > Sc$
 (c) $Zn > Cu > Ni > Sc$
 (d) $Sc > Zn > Ni > Cu$
136. The property exhibited by only transitional elements
 (a) To form paramagnetic compounds
 (b) To form coloured compounds
 (c) To form complex compounds
 (d) To show inert tendency
137. Which of the following will have standard oxidation potential less than *SHE*
 (a) *Zn* (b) *Cu*
 (c) *Fe* (d) *Ni*
138. Hydrated Cu^{+2} ion will be
 (a) Green (b) Violet
 (c) Blue (d) Colourless
139. The placement of *Zn*, *Cd* and *Hg* along with '*d*' block elements is not proper because
 (a) Their '*d*' orbitals are completely filled
 (b) Their '*d*' orbitals are empty
 (c) They do not form complex compounds
 (d) They do not form coloured compounds
140. Which of the following is the weakest reducing agent
 (a) *Zn* (b) *Cu*
 (c) H_2 (d) *Li*

